



Poster Abstract: LLM-Slice: Dedicated Wireless Network Slicing for Large Language Models

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Abstract

The rapid adoption of large language models (LLMs) presents new challenges for existing network architectures due to significant peak traffic and high communication uncertainty. Traditional wireless networks struggle to support efficiently, leading to intolerable response delays, disconnections, and resource wastage. To address these issues, we propose *LLM-Slice*, the first system to provide dedicated communication slices for LLMs within a wireless network environment. By creating LLM-specific network slices, LLM-Slice efficiently binds services with communication resources. Based on user equipment (UE) requests and a permissions database, the system registers specific slices to offer controllable LLM services, integrating a downlink resource control module to optimize response speed, enhance resource utilization, and reduce disconnections. By deploying and validating in a real UE-gNB-CN environment, numerical results demonstrate that LLM-Slice significantly improves response speed and resource efficiency, providing a novel solution for fast and controllable LLM access in wireless networks.

CCS Concepts: • **Network** → Network architectures.

Keywords: LLM, Network Slicing, Wireless Communication

1 Introduction

In recent years, the rapid development and widespread application of LLMs have been revolutionizing the field of artificial intelligence. From natural language processing to complex problem-solving, LLMs have demonstrated unprecedented capabilities in text understanding and generation. However, wireless communications have become one of the bottlenecks to accessing LLM services, primarily due to the special

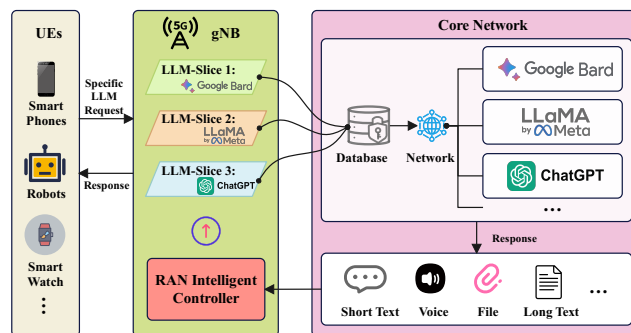


Figure 1. The Overview of the LLM-Slice System

traffic patterns and resource demands of LLMs. Traditional wireless network architectures face severe challenges: 1) LLM applications often generate **significant traffic peaks**, leading to network congestion and degradation of service quality; 2) LLM-generated responses vary greatly in length and complexity, leading to **high downlink communication uncertainty**, making efficient allocation of network resources difficult; 3) Due to improper allocation of network resources, users may experience noticeable **response delays**, affecting user experience; 4) When processing large amounts of LLM-generated data, **downlink disconnections** may occur, resulting in information loss and service interruptions. Therefore, a lack of optimization for LLM traffic characteristics will lead to inefficient resource utilization and performance degradation.

To address these challenges, we propose and implement *LLM-Slice*, the first 5G wireless network system to provide dedicated communication slices for LLMs. As shown in Figure 1, LLM-Slice creates LLM-specific network slices to efficiently bind services and communication resources. At first, UEs request specific LLM services from gNB where LLM-Slice implements multiple dedicated slices (e.g., Google Bard, LLaMA, ChatGPT), allowing different LLM services to coexist with independent resource allocation and management. Then, the core network includes a permission database and a control module for managing user access and resource allocation. At last, a RAN Intelligent Controller (RIC) dynamically adjusts wireless resources based on LLM response characteristics, enhancing system adaptability and optimizing network resource utilization.

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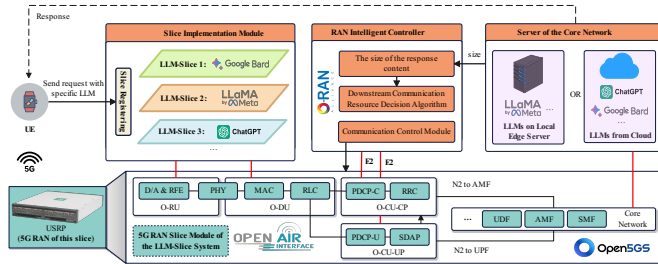


Figure 2. The Framework of the LLM-Slice System

2 System Design

Figure 2 shows the system architecture, which consists of three fundamental components: slice implementation, RIC, and a core network server. This design enables efficient network slicing management to support various LLM services.

Slice implementation. This module has a multi-tier architecture with three layers: the service layer, the network function layer, and the infrastructure layer. Each layer has specific tasks that contribute to slice deployment and implementation. The service layer manages LLM-specific slices, while the network function layer handles resource allocation and the infrastructure layer provides the physical resources.

RAN intelligent controller. This module dynamically optimizes resource allocation based on the characteristics of LLM responses. This intelligent system analyzes content size and employs sophisticated algorithms to make real-time decisions on resource allocation, ensuring optimal network performance for LLM services. It adapts to varying LLM traffic patterns, enhancing system flexibility and efficiency.

Core network server. It includes a permissions database to authenticate and authorize users for specific LLM services. Additionally, it hosts the communication control module, which interfaces with the RIC to implement resource allocation decisions and manage the end-to-end network slices for LLM services.

The system operates through a coordinated workflow. Initially, the UE sends an LLM service request to the gNB. The slice implementation module then registers the appropriate LLM-specific slice based on the request. The core network server verifies user permissions and activates the slice. As the LLM generates responses, the RIC dynamically adjusts resource allocation based on response characteristics. This process ensures efficient data transmission from the LLM to the UE through the optimized network slice.

3 System Implementation

We implemented the LLM-Slice system on a real-world 5G testbed, as depicted in Figure 3. The platform consists of several key components: a host machine running the Core Network functionality, connected to an Edge Server acting as a computing module; a USRP X410 software-defined radio device serving as the gNB for LLM-Slice; and a PC equipped

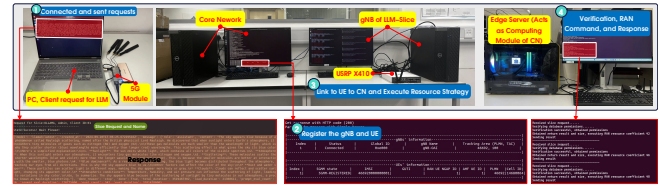


Figure 3. Real-World Implementation of LLM-Slice

with a 5G module functioning as the client to send LLM requests. These components are interconnected via a 5G network, collectively realizing the functionality of LLM-Slice.

1) **For the RAN**, we utilized a USRP X410 software-defined radio programmed with a modified version of the OpenAir-Interface [1] gNB software. We implemented LLM-specific slice support in the MAC layer and added interfaces to communicate slice information with the core network. 2) **The UE**, represented by a PC with a 5G module, was configured to request LLM-specific slices. We developed a client application that sends LLM requests and measures response times. 3) **For RIC**, we implemented the RIC as a separate process on the Edge Server. This controller interfaces with both the gNB and core network using the O-RAN E2 interface, which we extended to support LLM-specific metrics. 4) **For LLM integration**, we deployed the LLaMA model on the Edge Server. We utilized a custom API to handle requests from the core network and return responses, which are then transmitted through the optimized network slice. The overall operating status is illustrated in Figure 3.

4 Preliminary Results

Table 1. Comparison of LLM-Slice and Baseline 5G

Metric	Baseline	LLM-Slice	Improv.
Avg. Latency	250 ms	120 ms	52%
Resource Utilization	65%	85%	30.8%
Downlink Stability	92%	99%	7.6%

The numerical results in Table 1 show promising improvements across all metrics. LLM-Slice achieved a 52% reduction in average response latency. Resource utilization improved by 30.8%, indicating more efficient network management. Downlink stability also saw a notable increase, potentially reducing disconnections during LLM interactions. While these preliminary findings are encouraging, further studies are needed to evaluate LLM-Slice’s performance with more samples and with different LLM workloads.

References

- [1] Navid Nikaein, Mahesh K Marina, Saravana Manickam, Alex Dawson, Raymond Knopp, and Christian Bonnet. 2014. OpenAirInterface: A flexible platform for 5G research. *ACM SIGCOMM Computer Communication Review* 44, 5 (2014), 33–38.